

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Code:	CSA16	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	10% of CIA	Total Marks:	100 (5 Questions × 20 Marks)
CO1 Statement:	Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2		
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Section / Batch:		Date:	

Instructions to Students

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Short Answer Test	Architecture of Data Warehouse, ETL, OLAP	Q1 – Q5	Q1 –20 Q2 –20 Q3 –20 Q4 –20 Q5 –20
GRAND TOTAL:			100 Marks

SECTION A – SHORT ANSWER TEST

1. Suppose that a data warehouse consists of the three dimensions time, doctor, and patient, and the two measures count and charge, where charge is the fee that a doctor charges a patient for a visit.

(a) Enumerate three classes of schemas that are popularly used for modelling data warehouses.

(b) Draw a schema diagram for the above data warehouse using one of the schema.

(c) To obtain the same list, write an SQL query assuming the data is stored in a relational database with the schema fee (day, month, year, doctor, hospital, patient, count, charge). (5 Marks)

2. Suppose that a data warehouse for Big University consists of the following four dimensions: student, course, semester, and instructor, and two measures count and avg grade. When at the lowest conceptual level (e.g., for a given student, course, semester, and instructor combination), the avg grade measure stores the actual course grade of the student. At higher conceptual levels, avg grade stores the average grade for the given combination. (5 Marks)

(a) Draw a snowflake schema diagram for the data warehouse.

(b) Starting with the base cuboid [student, course, semester, instructor], what specific OLAP operations (e.g., roll-up from semester to year) should one perform in order to list the average grade of CS courses for each Big University student.

(c) If each dimension has five levels (including all), such as “student < major < status < university < all”, how many cuboids will this cube contain (including the base and apex cuboids)?

Data Warehouse Schema.

The three popular Schemas used in data Warehousing are Star Schema, Snowflake and fact Constellation Schema. For the given datawarehouse, we can use a Star Schema where Time, Doctor and Patient are the dimension tables connected to a central fact table. The fact table contains the two measures Count and Charge.

The relational database contains the tables fee (day, month, Year, doctor, hospital Patient, Count, Charge). The SQL query to retrieve the data is `SELECT day, month, Year, doctor, hospital, Patient, Count, Charge FROM fee;`

2 Big University Data Warehouse.

A Snowflake Schema for Big University Contains four dimensions: Student Course

Semester and Instructor, with count of
Average Grade as Measures in the fact
table. To find the average grade of
CS course for each Student, we first
select the CS course using a slice
or slice operation, then roll up the
required dimensions and group the
result by Student to calculate the
average grade. if each of the four
dimensions has five levels, the total
number of cuboids is $5 \times 5 \times 5 \times 5 = 625$
cuboids, including the base and apex
cuboids.

3) Weather Data Warehouse:

The weather data warehouse is
designed to store data collected from about

to get probes every hour for more than

of 1m years. The Main dimension are Time

and location, where Time contains levels

Such as days, day, month and year

and location contains probe and region

details. The Fact table store the measures

air pressure, temperature, and precipitation

OLAP Operations Such as roll-up, drill

down, Slice and dices can be used

to analyse weather conditions and

Identify general weather patterns based

on time and location.

4 Data Cube:

A data cube has n dimensions

and each dimension has p distinct values.

in the base Cuboid. The Maximum number

of cells in the base Cuboid is p^n , while the minimum number of cells is 1.

Since there are no concept hierarchies search dimension has a base level and an ~~ALL~~ level. Therefore the maximum number of cells in the complete data cube is $(p+1)^n$, and the minimum number of cells in the complete data cube is 2^n .

5) University Academic Performance Data
warehouse.

The University Academic Performance data warehouse uses ~~Student~~, course

~~Faculty~~ and Time as dimension. The central Fact table contains the measures

Total marks and attendance percentage. To display average marks department-wise for each academic year, we roll up the Student dimension to Department and the Time dimension to Year.